Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
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Ecosystem Evolution: A Stochastic Cellular Automaton Approach to Forest-to-Farm

Summary

As global population growth and increasing land resource demands drive the conversion of forests into agricultural land, significant changes occur in ecosystems, particularly in **biodiversity**, **soil quality**, and **hydrological cycles**. This study uses mathematical modeling to quantitatively analyze the dynamic changes in ecosystems during forest-to-farm conversion and explores the impact of various agricultural practices on ecosystem **stability** and **sustainability**.

First, we construct an agricultural ecosystem model based on a **food web framework**, incorporating agricultural cycles and seasonal variations. The study examines the effects of **herbicide** and **pesticide** use on plants, insects, and the overall ecosystem. A **predator-prey differential equation model** is developed to explore the mechanisms through which different agricultural practices influence ecological balance. Additionally, a **stochastic cellular automaton model** is introduced to simulate the long-term effects of chemical interventions on species distribution and ecosystem evolution. **The dynamic simulation results** indicate that while the moderate use of chemicals aids in **pest control** and **plant growth**, excessive use leads to **ecosystem imbalance**, manifesting as **reduced plant diversity** and **deteriorating soil quality**.

Next, the study investigates the role of **species reintroduction**, focusing on **pioneer trees** and **piper species** in border habitats, in restoring agricultural ecosystems. By refining the rules of the stochastic cellular automaton, the simulations reveal that different parameter settings lead to various ecological states following species reintroduction. **Furthermore, equilibrium point analysis** validates the accuracy of the model, showing that species reintroduction can lead to **five distinct stable states**, including **complete coexistence** or **predator-prey limit cycle states**.

Subsequently, the research explores the long-term impacts of different agricultural practices (reduced herbicide use and the introduction of key species) on ecosystem stability and biodiversity. Adjustments to the stochastic cellular automaton model **quantify the effects** of these practices on ecosystem recovery and long-term stability. Results demonstrate that **reducing herbicide use** alters the competitive dynamics between plants and herbivores, while **introducing bats** significantly improves **pest control efficiency** and **plant reproduction rates**, thereby enhancing ecosystem stability. Additionally, **rhizobia**, as plant symbionts, play a significant role in improving **soil fertility** and promoting **plant growth**.

Finally, the study further evaluates the **ecological benefits** of various organic farming practices, focusing on their comprehensive impacts on **pest control**, **crop health**, **biodiversity**, **long-term sustainability**, and **cost-effectiveness**. Quantitative evaluation metrics are proposed to compare the advantages of **different organic farming scenarios**.

The study also includes **sensitivity analysis** and an assessment of the model's **strengths** and **weaknesses**. The findings offer valuable insights for advancing **sustainable agricultural development**.

Keywords: Agricultural Ecosystem, Food Web Model, Predator-Prey Differential Equation Model, Stochastic Cellular Automaton, Dynamic Simulation

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1 Introduction

1.1 Problem Background

As the global population continues to grow and the demand for land resources increases, forests are being extensively cleared for agricultural development, resulting in significant ecological changes. Once lush and vibrant, these forests were sources of biodiversity. However, with the expansion of agriculture, the previously balanced ecosystems have been disrupted, replaced by monocultural farming practices. This shift has led to soil degradation, pest outbreaks, and significant challenges to agricultural productivity.

In response to these challenges, farmers have increasingly relied on chemical fertilizers and pesticides, though these measures often exacerbate the imbalance of the land's ecology. In these human-dominated agricultural ecosystems, the return of species such as bats and birds is considered a critical indicator of ecological recovery. However, ecosystem restoration is a prolonged process that requires gradual recovery of biodiversity, ultimately achieving sustainable agricultural development, as shown in Figure 1.

Figure 1 Making room for agriculture: challenges and restoration

1.2 Problem Background

In the Consideration of Mature Agricultural Practices (COMAP) group, our task is to select a specific region, collect data, and develop a mathematical model that tracks habitat changes from forests to farmlands, incorporating both natural processes and human decisions.

Task 1: This task aims to build a food web model for agricultural ecosystems, considering agricultural cycles and seasonality. It also explores the impact of chemical substances on ecosystem components and their role in disrupting system balance.

Task 2: As edge habitats mature, species return becomes key to restoring ecological balance. This task analyzes the interactions between species and their environment, assessing their effect on ecosystem stability.

Task 3: This task examines the impact of agricultural practices, such as herbicide removal and species introductions, on ecosystem stability, evaluating how these factors either support or disrupt ecological balance.

Task 4: his task investigates the effects of organic farming on biodiversity, crop health, and soil fertility, exploring its long-term sustainability and economic viability.

1.3 Literature Review

Agricultural ecosystem modeling is a core tool for understanding the interactions between agricultural production and the natural environment. Food web models [1], differential equations [2], and cellular automata model [3] have been widely applied in this field to describe species interactions, population dynamics, and spatial distribution processes.

Food web models simulate the flow of energy and species relationships within ecosystems, revealing the potential impacts of agricultural activities on ecological balance [4]. As a form of differential equations, predator-prey models analyze species population dynamics and provide a quantitative understanding of the impact of predation on agricultural ecosystem stability [5]. Cellular automata models, by simulating land use changes and species migration processes, reflect the spatial distribution patterns within agricultural ecosystems [6].

These models offer a multidimensional perspective for dynamic analysis of agricultural ecosystems, especially in evaluating the effects of agricultural interventions—such as the use of chemicals and species introduction—on ecosystem stability and biodiversity [7].Stochastic cellular automata models, with their ability to simulate complex ecological processes, particularly in capturing randomness and local interactions, provide a more realistic representation of species spatial distribution and evolutionary processes in agricultural ecosystems[8].

Furthermore, these models have been extensively applied in green agriculture research, particularly in assessing the long-term impacts of organic farming on ecosystems [9]. By simulating changes in biodiversity, crop health, and soil fertility under organic farming, these models offer theoretical support for sustainable agricultural development.

1.4 Our work

Task 1: Based on the Forest-to-Farm food web, we aim to simulate the interactions between producers and consumers within the ecosystem, incorporating agricultural cycles and seasonal variations. A predator-prey differential equation-based food web model will be constructed to analyze the effects of herbicide and pesticide use on plants, insects, and ecosystem stability. To enhance the model's realism, we will develop a stochastic cellular automata-based food web model for dynamic simulation, tracking ecosystem changes and accurately modeling the spatial distribution of species and the long-term impacts of chemical interventions.

Task 2: The task focuses on analyzing the processes of two species returning to edge habitats and their interactions with the existing agricultural ecosystem. The stochastic cellular automata model will be refined to evaluate the effects of species return on ecosystem stability, species diversity, and ecological processes. Dynamic simulations will be used to model different scenarios following species return, and the model’s accuracy will be validated through equilibrium stability analysis of the differential equations.

Task 3: This task examines the impact of different agricultural practices on ecosystem stability, focusing on measures such as herbicide removal and bat introduction to restore ecological balance. The stochastic cellular automata model will be adjusted, and its parameters

will be calibrated to quantify the effects of these practices on long-term ecosystem stability and biodiversity. For example, herbicide removal alters the competitive dynamics between plants and herbivores, while bat introduction enhances pest control and plant reproduction.

Task 4: This task aims to analyze the impact of organic farming under different scenarios on the entire ecosystem and its components. It will address issues such as pest control, crop health, plant reproduction, biodiversity, long-term sustainability, and cost-effectiveness. The model will be modified to reflect different organic farming scenarios, and evaluation metrics will be established to quantitatively compare the advantages of organic farming in each scenario.

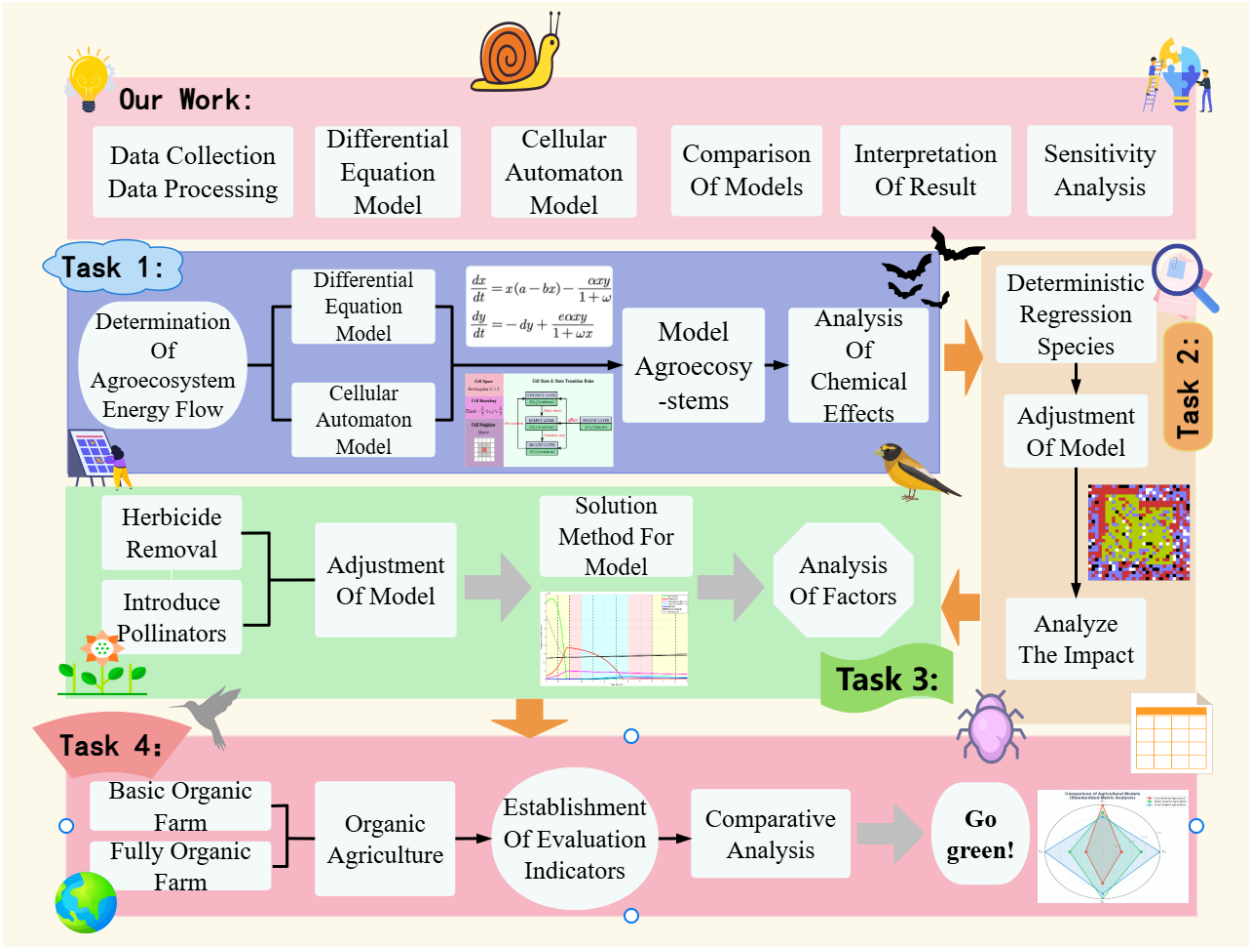


Figure 2 Our work

2 Assumptions and Justifications

Assumption 1: The impact of chemical agents on consumers in the agricultural ecosystem is ignored.

Justification: Chemical agents have a minimal short-term impact on the ecosystem, and their toxic effects accumulate slowly over time. Thus, the influence on consumers such as insects, birds, and bats is not considered in the model.

Assumption 2: Insect resistance to pesticides is ignored.

Justification: Insects develop resistance to pesticides over time, but this process occurs slowly. Therefore, the model assumes no significant change in pesticide effectiveness during the timescale of the study.

Assumption 3: The natural loss of resources in the agricultural ecosystem is ignored.

Justification: Natural resource loss (e.g., evaporation and soil erosion) is minimal and difficult to quantify accurately. The model simplifies by assuming that inorganic resources in farmland are only lost through absorption by producers.

Assumption 4: Interactions between producer and consumer populations are local.

Justification: Producers interact only with nearby consumers, and no direct interactions occur with distant populations. This simplifies the model and reflects localized ecological interactions.

3 Notations

The primary notations used in this paper are listed in Table 1.

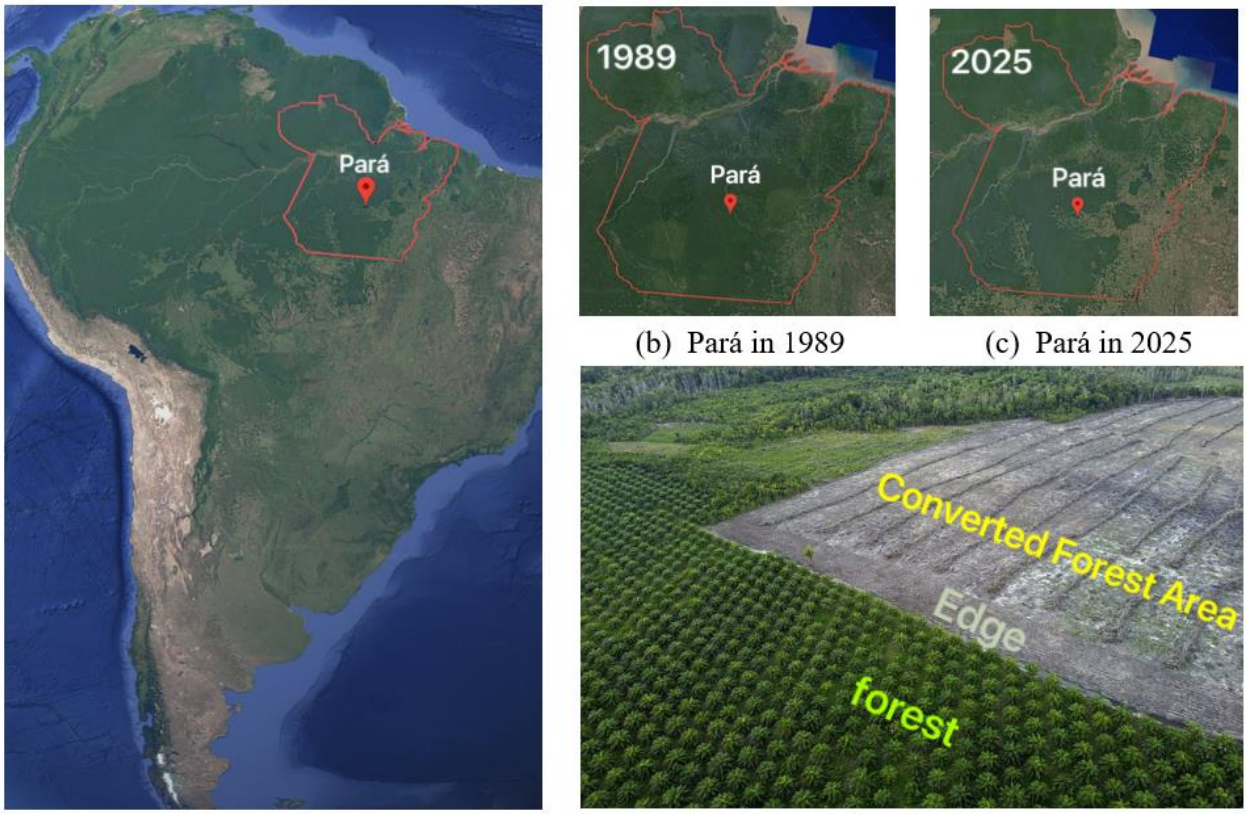
Table 1 Notions and Description

Symbols	Description	Units
α_i^j	attack efficiency of class i consumer j season	/
h_i^j	i consumer j seasonal digestion processing time coefficient	/
β_p	maximum conversion rate of resources absorbed by producers	/
k_p	producer's half packet and proportion	/
β_{c_i}	conversion rate of consumer i to food	/
β_r	conversion rate of organic matter to inorganic matter by decomposer	/
h_e	herbicide inhibitor	/
p_e	insecticide impact factor	/
t_0	natural recovery time	/
t_p	processing time	/
t_m	edge zone ripening time	/
$\xi_p^{'}$	reproduction probability of the producer in the presence of bats	/
I_i	i seasonal resource input	/

4 Model Preparation

4.1 Overview of the Research Area

The study area for this research is located in the state of Pará, Brazil, between 1°27'21" S and 1°45'47" S latitude, and 48°30'14" W and 50°00'16" W longitude. Pará has a tropical rainforest climate, with an average annual temperature of approximately 26°C and annual precipitation exceeding 1500 mm. The region is predominantly covered by tropical rainforest vegetation, with vast forest areas. In recent years, as agricultural development has progressed, large areas of forest have been cleared and converted into agricultural land, particularly for the cultivation of soybeans, palm oil, and other cash crops. This transformation has led to significant ecological impacts, especially in terms of biodiversity, soil quality, and the hydrological cycle. Figure 3 shows the geographical location of Pará and the distribution of agricultural land following deforestation.



(a) Location of Pará in South America (b) Pará in 1989 (c) Pará in 2025 (d) Forest-to-farm in Pará

Figure 3 Forest-to-Farm Transition in Pará: satellite observations and land use

Figure 3(a) shows the geographical location of Pará in South America. Figures 3(b) and 3(c) present satellite images of Pará in 1989 and 2025, respectively. From the images, it is evident that certain areas have been converted into agricultural land, with a significant reduction in the coverage of tropical rainforests. Figure 3(d) illustrates an example of forest-to-farm conversion in Pará, where the land is divided into three main regions, showing different stages of deforestation and subsequent land-use changes. Selecting Pará as the study area is of significant scientific importance and provides valuable data support for global ecological conservation and sustainable agricultural development.

4.2 Data Overview

The data for this study is sourced from several authoritative institutions, covering a wide range of information such as climate, vegetation, and geographical location, providing reliable and accurate data support for the research. The data sources are listed in Table 2.

Table 2 Institution, Data Website and Data Type

Institution	Data website	Data Type
Google	https://earth.google.com/	Location
EPAE	https://resources.unep-wcmc.org	Climate
Brazil government	https://ibge.gov.br/en/home-eng.html	Population in Pará
Brazil Environment Agency	https://www.gov.br/mma/pt-br	Climate in Pará
Chinese government	https://www.ccpit.org/	Climate in Pará

5 Task 1: Forest-to-Farm Ecosystem Modeling

5.1 Food Web Model of the Ecosystem

The food web is a complex network of feeding relationships between organisms in an ecosystem. It illustrates how energy and nutrients flow and interact within an ecosystem, from producers P to various types of consumers $C_1, ..., C_n$, and finally to decomposers D. Pará, Brazil, is an ideal example of the Forest-to-Farm process. According to the literature [10], we can define the food web structure and energy flow during the Forest-to-Farm process in this region, as shown in Figure 4:

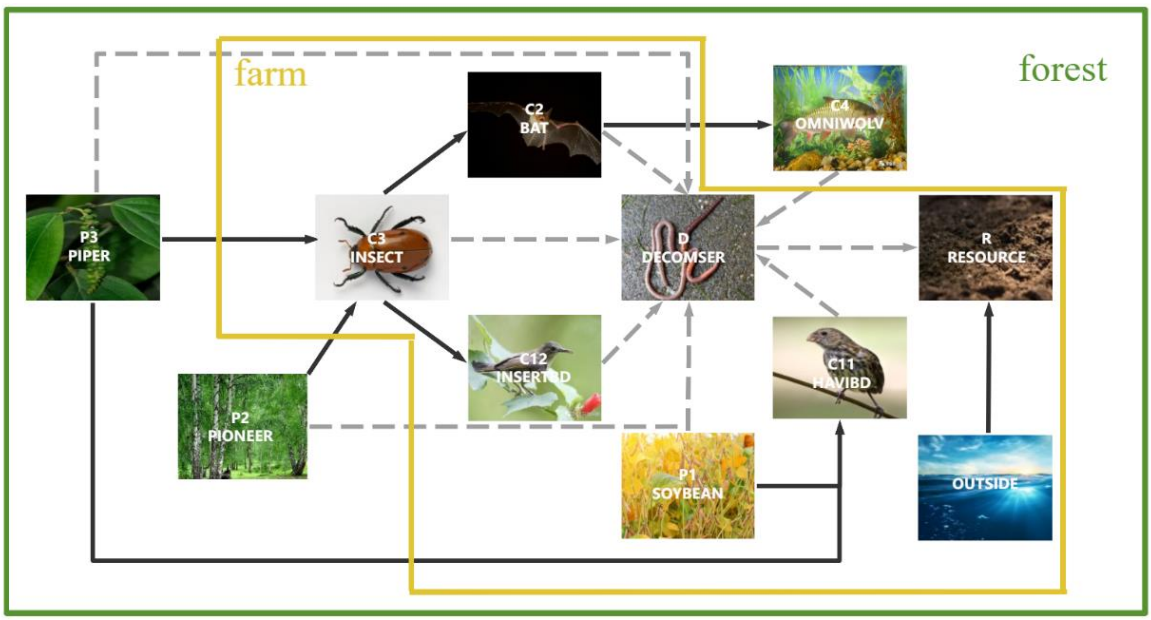


Figure 4 Food Web Model structure and energy flow of Forest-to-Farm Ecosystem

Figure 4 shows two food web models: the food web model of the agricultural ecosystem (highlighted in yellow), which serves as the foundation and focus for our subsequent modeling work, and the food web model of the forest ecosystem (highlighted in green), which is in a stable state prior to human intervention.

In the agricultural ecosystem, the primary producer is soybean P_1, with main consumers including birds C_1 such as herbivorous birds C_{11} and insectivorous birds C_{12}, bats C_2, and insects C_3 [11]. Decomposers D are microorganisms, and resources R include sunlight, water, and inorganic salts. In the forest ecosystem, in addition to the organisms found in the agricultural food web, it also includes unique producers from the Brazilian tropical rainforest[12], such as pioneer tree P_2 and pepper species P_3, along with other consumers C_4.

Through the energy flow diagram laying the groundwork for subsequent differential equation and stochastic cellular automata modeling.

5.2 Differential Equation-Based Food Web Model

5.2.1 Introduction to the Differential Equation Model

This section focuses on the competitive relationships among populations in the agricultural ecosystem food web model. Based on the energy flow diagram, a predator-prey and competition

differential model was constructed to analyze the dynamic characteristics of populations and their evolution under mutual interactions. Referring to the relevant studies in [13], a predator-prey model incorporating a Holling Type II functional response was developed, tailored to the population characteristics of the study area:

$$\begin{cases} \dfrac{dx}{dt} = x(a - bx) - \dfrac{\alpha xy}{1 + \omega x} \\ \dfrac{dy}{dt} = -dy + \dfrac{e\alpha xy}{1 + \omega x} \end{cases} \tag{1}$$

5.2.2 Differential Equation Model Construction

Given that the predator and prey populations in this system are numerous and their interactions are complex, we model each predator-prey relationship based on the energy flow diagram and derive the differential equations for the complete ecosystem:

$$\begin{cases} \dfrac{dp_1}{dt} = k_{p_1} p_1 \left(1 - \dfrac{1}{k} p_1\right) - c_{11}\phi(p_1) - c_{31}\phi(p_1) \\ \dfrac{dc_3}{dt} = c_3[-k_{c_3} + \beta_{c_3}\phi(p)] \\ \dfrac{dc_{11}}{dt} = c_{11}[-k_{c_1} + \beta_{c_1}\phi(p)] \\ \dfrac{dc_{12}}{dt} = c_{12}[-k_{c_{12}} + \beta_{c_{12}}\phi(c_{31})] \\ \dfrac{dc_2}{dt} = c_2[-k_{c_2} + \beta_{c_2}\phi(c_{31})] \\ \dfrac{dd}{dt} = [k_c(c_{11} + c_{12}) + k_2 c_2 + k_3 c_3]\dfrac{d}{1 + h_d d} - k_d d \\ \dfrac{dr}{dt} = I - \dfrac{\beta_p r}{k_p + r} p + \beta_r d[k_{c_1}(c_{11} + c_{12}) + k_{c_3} c_3 + k_{c_2} c_2] \end{cases} \tag{2}$$

This system of differential equations integrates the dynamic evolution between predator-prey pairs, providing an accurate mathematical description for subsequent research. In the equation, p, c, d, s, r represent the populations of producers, consumers, decomposers, and resources, respectively; $\phi(\cdot)$ denotes the Holling Type II functional response; represents the mortality rate for each component of the food web; β_{c_i} indicates the conversion rate of consumers to decomposers; and β_r denotes the conversion rate of decomposers to resources. I represents a correction factor that accounts for the subsequent agricultural cycle and seasonal variations, which will be discussed in detail later. It is worth mentioning that to model the limiting effect of substrate concentration on growth rates, we consider using the Monod equation: $\beta_p r / k_p + r$ which describes the relationship between microbial growth rates and substrate concentration .

The differential equations describe the complex food web system: Producers P provide the energy base through their growth and resource utilization, supplying energy to consumers C for predation. The population of consumers C is influenced by predation rate, mortality rate, and predation intensity. Decomposers D break down organic matter, converting nutrients into inorganic resources that are available for producers, thus promoting the growth and reproduction of producers.

5.2.3 Modifications to the Differential Equation Model

Although the differential equations in Equation (2) describe species interactions within the agricultural ecosystem, they fail to account for agricultural cycles and seasonal variations. Thus, the model requires modifications to include periodic and seasonal factors, enhancing its accuracy and predictive capability in representing real-world agricultural environments.

In Pará, Brazil, the "Forest-to-Farm" process utilizes the dry season for land clearing, the rainy season for growth, and the transition period for harvesting, achieving efficient resource utilization. The modeling process incorporates time, climate, and crop conditions to optimize agricultural production, as illustrated in Figure 5.

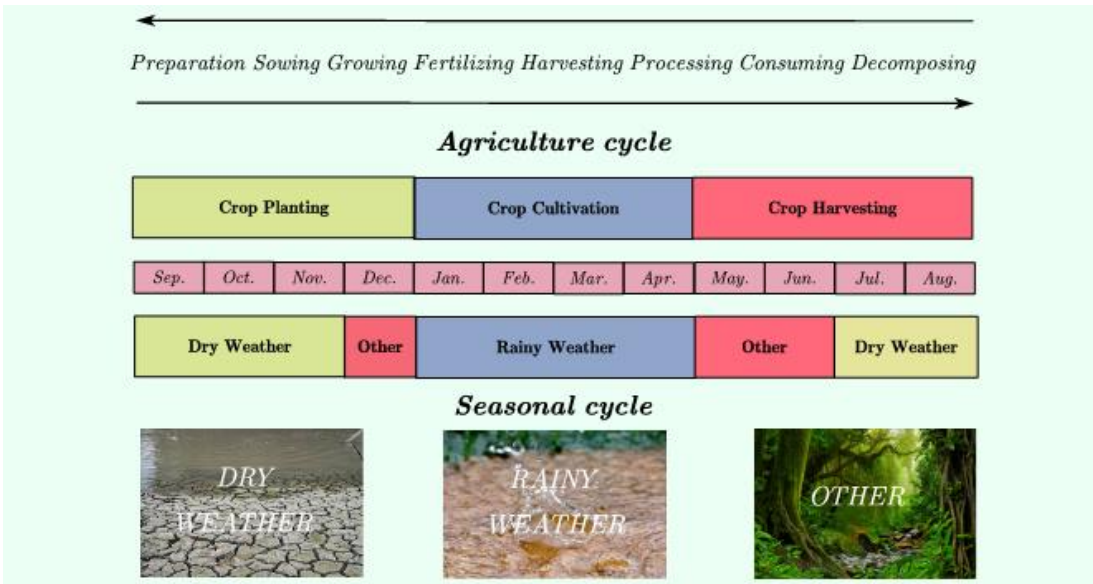


Figure 5 Process of Forest-to-Farm in Pará, Brazil

The study area experiences three main seasons: the rainy season (December to January), the dry season (September to October), and a normal season. These seasonal variations significantly influence the ecosystem, particularly in terms of natural factors such as water availability, sunlight, and temperature. These factors directly or indirectly impact plant growth, animal activity, and resource availability. Hence, the model is adjusted to reflect these seasonal characteristics, improving its ability to simulate the dynamic evolution of agricultural ecosystems across different seasons.

To address these factors, modifications are applied to resource input I^i, digestion rate $h_{c_k}^i$, and the handling efficiency k_p and attack efficiency α_{c_k} in $\phi(x)$. The revised equation is expressed as:

$$I^i = \begin{cases} I^1, if\ rainy\ season \\ I^2, if\ dry\ season \\ I^3, otherwise \end{cases}, k_p^i = \begin{cases} k_p^1, if\ rainy\ season \\ k_p^2, if\ dry\ season \\ k_p^3, otherwise \end{cases}$$

$$h_{c_k}^i = \begin{cases} h_{c_k}^1, if\ rainy\ season \\ h_{c_k}^2, if\ dry\ season \\ h_{c_k}^3, otherwise \end{cases}, \alpha_{c_k}^i = \begin{cases} \alpha_{c_k}^1, if\ rainy\ season \\ \alpha_{c_k}^2, if\ dry\ season \\ \alpha_{c_k}^3, otherwise \end{cases} \tag{3}$$

Farmers' activities at specific times significantly impact the agricultural ecosystem. The agricultural cycle includes a series of processes such as soil preparation, sowing, growth, fertilization and pest control, harvesting, post-harvest processing, consumption, and decomposition. We need to model these processes in stages to refine the model, considering the use of herbicides and pesticides during pest control to describe their time-dependent effects at different stages.

Soil Preparation and Sowing: Farmers clear forest plants $P_1 P_2$ and drive away other animals C_4, followed by sowing crop seeds:

$$p_1(0)=0, p_2(0)=0, c_4(0)=0, p(t)=p_0 \tag{4}$$

Cultivation Process: During crop growth, farmers fertilize and spray herbicides and pesticides.

- **Modification to Resource Quantity by Fertilization**: The fertilization input I_h modifies the resource quantity r, reflecting the effect of fertilization on crop growth:

$$\frac{dr}{dt}=I+I_h-\frac{\beta_p r}{k_p+r}p+\beta_r d[k_{c_1}(c_{11}+c_{12})+k_{c_3}c_3+k_{c_2}c_2] \tag{5}$$

- **Modification to Producer Quantity by Herbicides**: Herbicides inhibit crop growth by reducing the rate at which crops absorb resources, represented by the inhibition factor $h_e(0<h_e<1)$. It is assumed that herbicides do not affect other organisms and have minimal short-term impact due to their longer active periods:

$$\frac{dp_1}{dt}=k_{p_1}h_e p_1\left(1-\frac{1}{k}p_1\right)-c_{11}\phi(p_1)-c_{31}\phi(p_1) \tag{6}$$

- **Modification to Consumer Quantity by Pesticides**: Pesticides impact insect mortality rates, represented by the factor $p_e(p_e>1)$, to suppress insect population growth. It is assumed that insect resistance to pesticides is not considered:

$$\frac{dc_3}{dt}=c_3[-k_{c_3}p_e+\beta_{c_3}\phi(p)] \tag{7}$$

In the subsequent analysis, we will focus on herbicides and pesticides to evaluate their impact on ecosystem stability. Excessive use of these chemicals may disrupt plant diversity, unbalance insect populations, and threaten the survival of bats and birds, ultimately undermining ecological balance.

Harvesting: Farmers harvest crops annually, introducing t_n to represent the natural recovery time of crops after harvesting:

$$p(t-t_n)=0 \tag{8}$$

Post-Harvest Processing and Consumption: Harvested crops are processed, and the remaining residues are removed from the farmland and decomposed by decomposers. The crop residue quantity I_r is introduced to modify the decomposer population, reflecting the effect of post-harvest processing on decomposers:

$$\begin{cases} \dfrac{dd}{dt}=\beta_d[k_c(c_{11}+c_{12})+k_2c_2+k_3c_3+I_r]\dfrac{d}{1+h_d d}-k_d d \\ I_r=p(t-t_n)(1-\theta) \end{cases} \tag{9}$$

5.2.4 Model Solution and Analysis

The solutions for the food web model based on differential equations, considering and not considering the impact of chemicals, are shown in Figure 6 and Figure 7, respectively. In the

figures, the red, blue, and yellow regions represent the dry season, rainy season, and normal season, respectively, while the black dashed lines indicate 7 steps of the agricultural cycle.

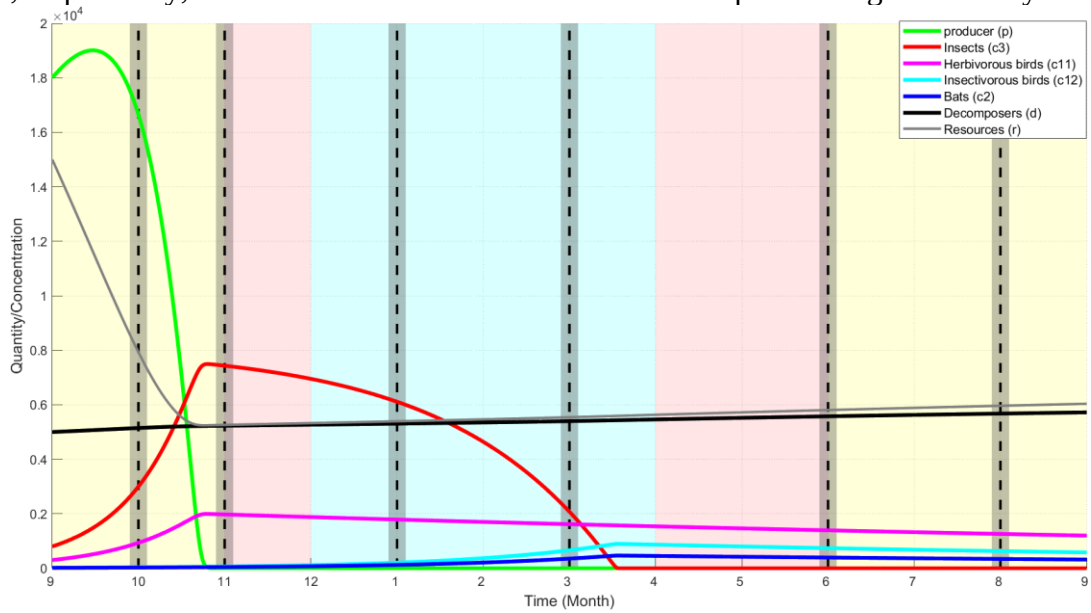


Figure 6 Solution results without considering the impact of chemicals

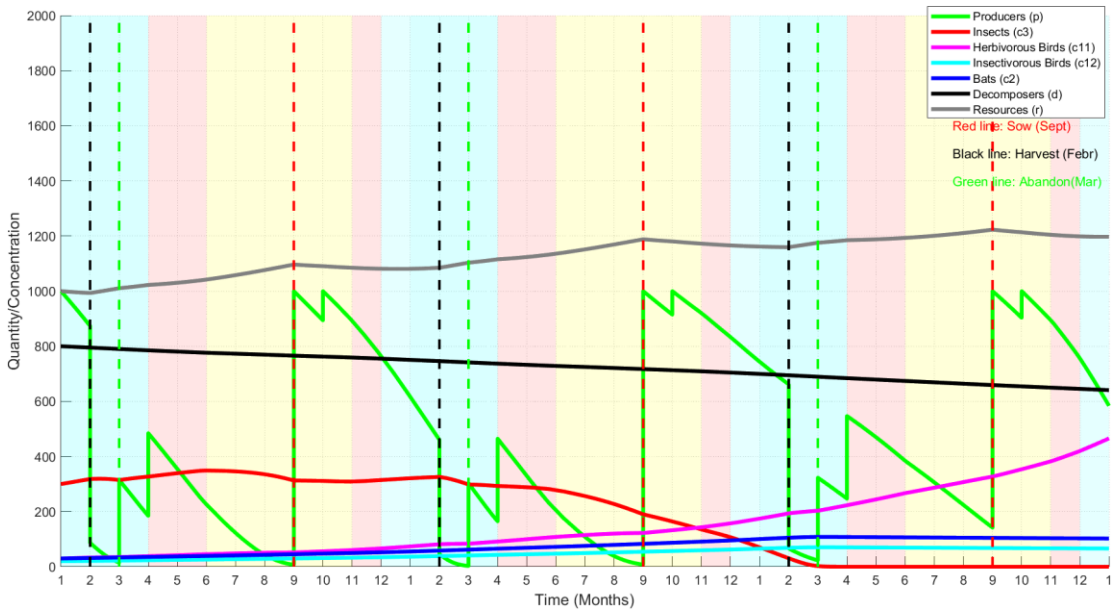


Figure 7 Solution results considering the impact of chemicals

From Figure 6, without chemicals, insects multiply rapidly, causing plants to decline and eventually leading to the collapse of the agricultural ecosystem.

From Figure 7, plants grow slowly during the dry season but accelerate during the rainy season, showing periodic growth after annual harvests. However, herbicides and pesticides, while controlling insects, indirectly harm insectivorous birds and bats, reducing soil fertility, biodiversity, and ecosystem stability.

In summary, excessive chemical use disrupts ecosystems, while moderate use balances pest control with ecological stability, ensuring sustainable agriculture and environmental health.

5.3 Cellular Automata-Based Food Web Model

5.3.1 Construction of the Stochastic Cellular Automata Model

Based on the food web model, a stochastic cellular automata model of the agricultural ecosystem was developed to simulate species interactions, energy flow, and resource distribution within the system, as illustrated in Figure 8.

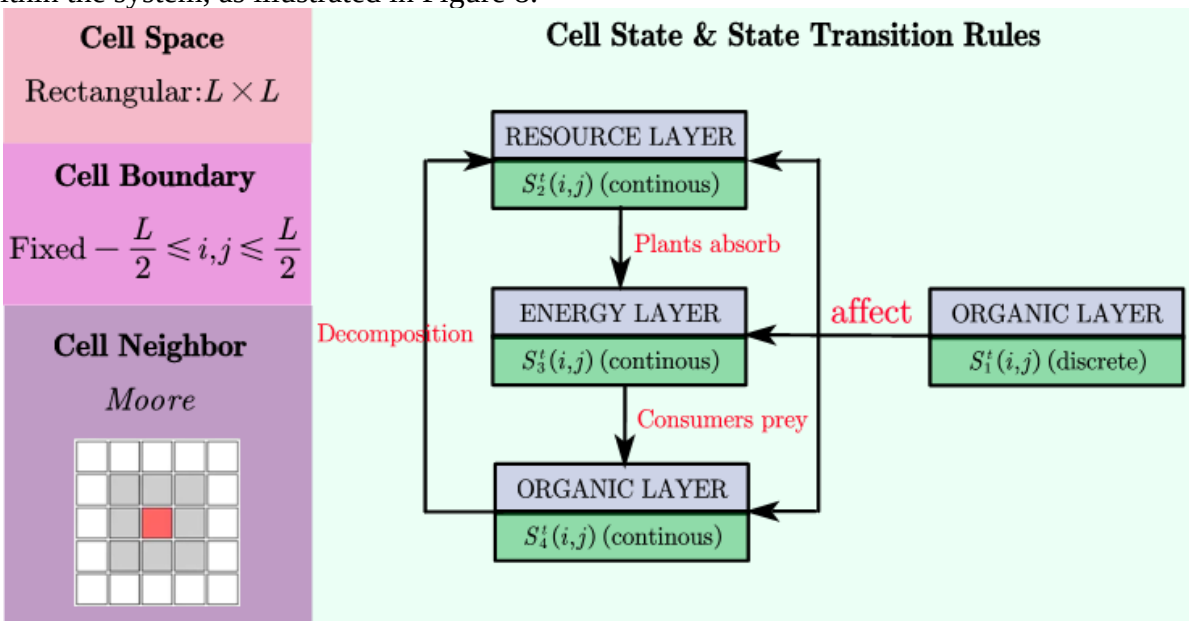


Figure 8 Stochastic Cellular Automaton Model

Cell Space: The cellular space of the model is a rectangular grid of size $L \times L$

Cell Boundary: The boundary conditions are fixed, meaning the state of each cell at the boundary does not reverse or propagate beyond the grid.

Cell Neighborhood: The neighborhood of each cell is the Moore neighborhood model.

Model Hierarchical Structure

- **Role Layer**: This layer describes the state of different ecological roles, representing the distribution and interactions of various species within the ecosystem. It is mathematically defined as $S_1^t(i,j) \in \{P, C_{11}, C_{12}, C_2, C_3, E\}$
- **Organic Matter Layer**: This layer describes the decomposition process of organic matter, representing the flow of organic material within the ecosystem. It is mathematically defined as $S_2^t(i,j) \in \mathbb{R}$
- **Resource Layer**: This layer represents the distribution of resources, specifically the amount of resources available to producers. It is mathematically defined as $S_3^t(i,j) \in \mathbb{R}$, representing the state of the resource layer at a specific cell.
- **Energy Layer**: This layer describes the energy reserves of each role, reflecting the flow and transformation of energy within the ecosystem. The storage and consumption of energy affect the survival and development of roles. It is mathematically defined as $S_4^t(i,j) \in \mathbb{R}$.

5.3.2 Stochastic Cellular Automata State Transition Rules

The state transition rules of the SCA simulate interactions between organisms and their environment by dividing the ecosystem into functional modules like the resource, energy, and organic matter layers. These rules define key behaviors such as resource absorption, reproduction, movement, and predation, influencing resource and energy flow. Table 3 highlights key rules for simplicity.

Table 3 Role Layer State Transition Rules (Partial)

Time t	Transition rule name	Time $t+1$
E	Producer reproduction	R
P	Bat movement or reproduction	$P+C_2$
C_3	Insectivorous bird predation	C_{12}
$P+C_2$	Bat movement or starvation death	P
$C_{11}+C_{12}$	Insectivorous bird movement or death	C_{11}

1. Resource Layer - Energy Layer Interaction Rules: When producers survive in the current grid, they absorb resources from the resource layer and convert them into energy.

Step 1: Check if there is a producer in the current grid, i.e., check if $S_3^t(i,j)=p$

Step 2: If a producer exists, the resource amount in the resource layer decreases:

$$S_3^t(i,j)=S_3^t(i,j)\,(1-\alpha_3)$$

Step 3: If a producer exists, the resources absorbed by the producer from the resource layer are converted into energy for the producer: $S_4^t(i,j)=S_4^t(i,j)+\alpha_3 S_3^t(i,j)$.

2. Reproduction Rules for Producers and Consumers: Organisms can reproduce when energy is sufficient, and new individuals will be born in adjacent empty spaces.

Step 1: Check if energy is greater than the reproduction, i.e., check if $S_4^t(i,j)>\theta_1$.

Step 2: Check if a random number is greater than the reproduction probability, i.e., check if $\xi^{(t+1)}(i,j)>\xi_p$

Step 3: Select a random Moore neighbor to reproduce: $n,m\in\{1,-1\}$.

Step 4: After reproduction, the offspring inherit the role and energy of the parent, and the parent's energy decreases:

$$S_1^{(t+1)}(i+n,j+m)=S_1^{(t)}(i,j)$$

$$S_4^{(t+1)}(i+n,j+m)=\alpha_1 S_4^{(t)}(i,j)$$

$$S_4^{(t+1)}(i+n,j+m)=(1-\alpha_1-\alpha_2)S_4^{(t)}(i,j)$$

3. Consumer Movement Rules: Consumers will move in the environment based on a certain probability.

Step 1: Check if there is a consumer in the current grid, i.e., check if $S_1^t(i,j)=c$

Step 2: Check if a random number is greater than the movement probability, i.e., check if

$$\xi^{(t+1)}(i,j)>\xi_m$$

Step 3: If conditions are met, select a random Moore neighbor to move: $n,m\in\{1,-1\}$, and update the original location to empty E:

$$S_1^{(t+1)}(i+n,j+m)=S_1^{(t)}(i,j)$$

$$S_4^{(t+1)}(i+n,j+m)=(1-\alpha_2)S_4^{(t)}(i,j)$$

4. Organism Survival Metabolism Rules: Each organism consumes energy to maintain life activities during its survival.

Step 1: Check if the current position is an organism. If it is an organism, its energy will be consumed according to the metabolism coefficient $S_4^{(t+1)}(i+n,j+m)=(1-\alpha_2)S_4^{(t)}(i,j)$

5. Energy Layer - Organic Matter Layer Interaction Rules: When the energy of an organism is below a certain threshold, it will die and transfer its energy to the organic layer.

Step 1: Check if energy is less than the death threshold, i.e., check if $S_4^t(i,j)<\theta_2$.

Step 2: If energy is depleted, the organism dies, and its original location becomes empty: E. After death, the organism's energy is transferred to the organic matter layer:

$$S_2^{(t+1)}(i,j) = S_4^{(t)}(i,j)$$

$$S_4^{(t+1)}(i,j) = 0$$

6. Consumer Predation Rules: Consumers will prey on other organisms under suitable conditions to gain energy.

Step 1: Check if there is a target organism to prey upon in the current grid.

Step 2: If the energy of the target organism is greater than the current organism's energy and they are adjacent, predation occurs. After predation, the prey's energy disappears, the predator's energy increases, and the prey's energy is transferred to the organic layer:

$$S_4^{(t+1)}(i,j) = 0$$

$$S_2^{(t+1)}(i+n,j+m) = S_2^{(t)}(i+n,j+m) + (1-\alpha_4) S_4^{(t)}(i+n,j+m)$$

$$S_4^{(t+1)}(i+n,j+m) = (1-\alpha_2) S_4^{(t)}(i,j) + \alpha_4 S_4^{(t)}(i+n,j+m)$$

7. Organic Matter Layer - Resource Layer Interaction Rules: The energy in the organic layer can be converted into resources in the resource layer, and vice versa.

Step 1: Organic matter transfers energy during decomposition

$$S_2^{(t+1)}(i,j) = S_2^{(t)}(i,j)\,(1-\alpha_5)$$

Step 2: The resource layer absorbs energy from the organic layer:

$$S_3^{(t+1)}(i,j) = S_3^{(t)}(i,j) + \alpha_5 S_2^{(t)}(i,j).$$

5.3.3 Model Solution and Analysis

The cellular automata simulation model was solved, and the population distribution in farmland under different treatment conditions is shown in Figures 9 and 10.

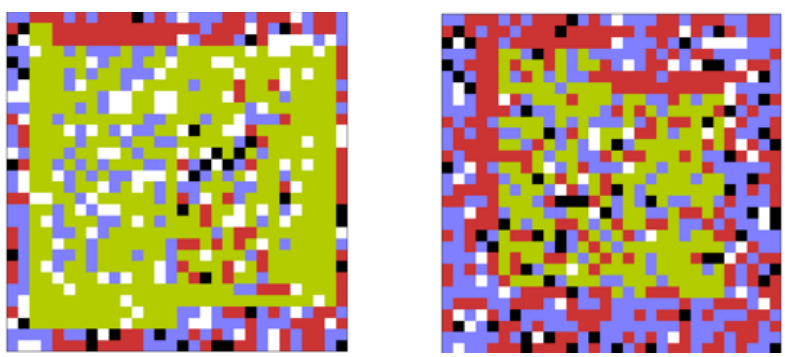


Figure 9 Population distribution results map without using chemicals

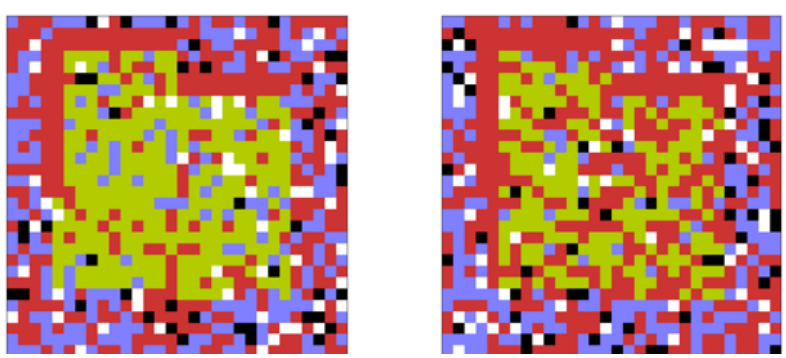


Figure 10 Population distribution results map using chemicals

Figures 9 and 10 show the population distribution over ten years, comparing fields with and without herbicides and insecticides. The results suggest that moderate use of these chemicals promotes healthy crop growth, with denser and more widespread plants. Proper chemical use helps maintain ecosystem stability, while without chemicals, the ecosystem is severely damaged, with crops facing heavy insect infestations within a month. This matches the predictions from the differential equation model, confirming the accuracy of the cellular automata model.

6 Task2: Species Regeneration in Ecosystems

6.1 Description of Species Regeneration

The health and stability of an ecosystem depend on the return of key species and the functional recovery of transitional zones. As a buffer between farmland and forest, the transitional zone connects two distinct ecosystems and plays an essential role in species dispersal, resource cycling, and biodiversity restoration. Studying species return and the maturation process of transitional zones helps us understand dynamic ecosystem changes and provides scientific support for future model adjustments.

Transitional Zone: The transitional zone refers to the area between farmland and forest, acting as a buffer zone. It typically has lower soil fertility and plays a crucial role in the species return process, connecting the two ecosystems. The range of this zone is between farmland $B \times B$ and forested areas $L \times L$, typically marked as $L < i,j < B$.

Species Regeneration: Species regeneration refers to the reintroduction and growth of specific plant or animal species in a given area under suitable conditions. In this study, pioneer trees P_2 and pepper species P_3 are chosen for detailed analysis. Pioneer trees are common in early ecological restoration stages, as they quickly colonize and restore land, while pepper species thrive in transitional zones and help promote biodiversity.

Maturation of Transitional Zones: Maturation of a transitional zone refers to the process in which the soil fertility gradually recovers and becomes suitable for plant growth. In this phase, native forest seeds naturally disperse and begin to germinate and reproduce. The maturation time t_m indicates how long it takes for the zone to reach a state capable of supporting diverse plant life, while p_s represents the number of native plant seeds in the zone, reflecting the potential for future biodiversity and ecological recovery.

6.2 Description of Species Regeneration

6.2.1 Modifications to the Model Hierarchical Structure

Role Layer State Adjustment: The role layer S_1 describes the status of various ecological roles, representing the distribution and interactions of species in the ecosystem. The role layer states have been expanded to the following set: $S_1^t(i,j) \in \{P_1, P_2, P_3, C_{11}, C_{12}, C_2, C_3, E\}$. Since the transitional zones include populations from the forest ecosystem P_2, P_3, the role layer has been adjusted to better reflect the species regeneration and ecological cycles in transitional zones and farmland.

6.2.2 Modifications to State Transition Rules

1. Resource Layer - Energy Layer Interaction Rule Modification: When producers are present in the current grid, they absorb resources from the resource layer and convert them into energy. With the introduction of new species, the rules are updated as follows:

Step 1: Check if there are producers in the grid, i.e., check if $S_3^t(i,j) = \{p_1, p_2, p_3\}$

Step 2: If producers exist, the resource amount in the resource layer decreases:

$$S_3^t(i,j) = S_3^t(i,j)\,(1-\alpha_3)$$

Step 3: If producers exist, the absorbed resources are converted into the producer's energy:

$$S_4^t(i,j) = S_4^t(i,j) + \alpha_3 S_3^t(i,j)$$

2. Supplementary Rules for External-Resource Layer Interactions:

The interaction rules for the resource layer describe the effects of external inputs on the grid state. These are divided into two categories: transitional zones and farmland.

- **Transitional Zone Interaction Rules**: Describe the impact of natural resource inputs on the land without considering human intervention: $S_3^{(t+1)}(i,j) = S_3^{(t+1)}(i,j) + I\,(only\ nature)$

  Transitional zone grid locations must meet the following condition: $\frac{B-L}{2} < |i|,|j| < \frac{L}{2}$

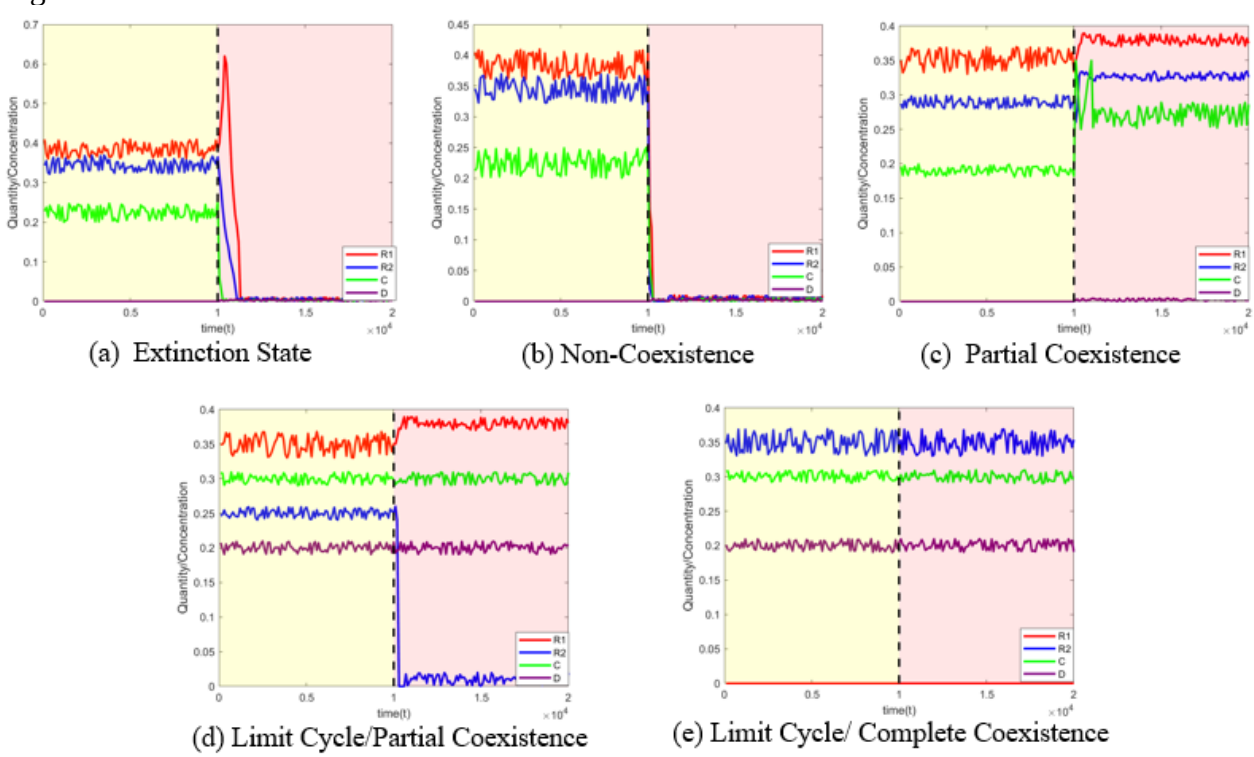
- **Farmland Interaction Rules**: Describe the effects of both natural resource inputs and human inputs, such as fertilization, on the land: $S_3^{(t+1)}(i,j) = S_3^{(t+1)}(i,j) + I + I_h$

  Farmland grid locations must meet the following condition: $|i|,|j| < \frac{L}{2}$

Other state transition rules remain unchanged.

6.2.3 Model Solution and Analysis

Through the analysis of the simulation results of the random cellular automata model, we observe that different outcomes may arise following species regeneration. Varying parameter settings can lead to different ecological states after species return. Under different parameter settings, the simulation shows several possible dynamic behaviors of the ecosystem, as shown in Figure 11:



(a) Extinction State (b) Non-Coexistence (c) Partial Coexistence

(d) Limit Cycle/Partial Coexistence (e) Limit Cycle/ Complete Coexistence

Figure 11 Solution results of the modified cellular automata model

Based on the different parameter settings, the system may reach an extinction state, a non-coexistence state, a partial coexistence state, or a predator-prey limit cycle. In some cases, species form stable coexistence states through competition and predation, while in other cases, the system may fall into species extinction or non-coexistence. These results highlight that the stability of the ecosystem is influenced by parameter variations, and appropriate ecological management and species control measures are critical to maintaining ecological balance.

6.3 Model Validation Based on Equilibrium Point Analysis

Simulations of the stochastic cellular automata model reveal varying outcomes after species regeneration, reflecting the inherent randomness of the model. To validate its accuracy, we analyzed the equilibrium points of the differential equation model, assuming sufficient mixing of populations. The results, shown in Table 4, align with the simulation outcomes in Figures 11(a)–(e), confirming the model's reliability in describing ecosystem dynamics.

Table 4 The results of equilibrium point analysis

Equilibrium Point	State Name	Dynamic Characteristics
$x_i = 0, i = 1, ..., n$	Extinction	No population growth, total extinction
$x_1 > 0, x_2, ..., x_n = 0$	Non-Coexistence	Resource-only system, susceptible to perturbations
$x_1, ..., x_3 > 0, x_4, ..., x_n = 0$	Partial Coexistence	Limited ecological interactions, simple dynamics
$x_1, ..., x_{n-1} > 0, x_n = 0$	Partial Coexistence or Limit Cycle State	Predator-prey dynamics, cycles or steady coexistence
$x_i > 0, i = 1, ..., n$	Complete Coexistence or Limit Cycle State	Full ecosystem dynamics, predator-prey oscillations

These results confirm that species regeneration can lead to varying ecosystem states depending on resource availability and species interactions.

7 Task3: Species Regeneration in Ecosystems

7.1 Modeling the Impact of Herbicide Removal on Ecosystems

To study the impact of herbicide removal on ecosystems, modifications to the cellular automata model are necessary. Herbicides, as widely used chemical agents, have significant effects on agricultural ecosystems, particularly on the growth and reproduction of plants, insects, and other organisms.

7.1.1 Modifications to State Transition Rules

To simulate the effects of herbicide removal on ecosystems, certain state transition rules in the cellular automata model need to be adjusted. The details are as follows:

1. Modification to Resource Layer - Energy Layer Interaction Rules: When producers are present in the current grid, they absorb resources from the resource layer and convert them into energy. With the removal of herbicides, the rules are modified as follows:

Step 1: Check if there are producers in the grid, i.e., check if $S_3^t(i,j) = \{p_1, p_2, p_3\}$.

Step 2: If producers exist, the resource amount in the resource layer decreases:

$$S_3^t(i,j) = S_3^t(i,j)\,(1 - \alpha_3)h_e$$

h_e indicates the inhibitory effect of herbicides on producers' ability to absorb resources from the resource layer.

Step 3: If producers exist, the absorbed resources are converted into the producer's energy:

$$S_4^t(i,j) = S_4^t(i,j) + \alpha_3 S_3^t(i,j)$$

Apart from these adjustments, other rules remain unchanged.

7.1.2 Model Solution and Analysis

Figure 12 illustrates the impact of herbicide removal on various components of the agricultural ecosystem. These changes indicate that the removal of herbicides alters the patterns of resource competition and biological interactions within the ecosystem.

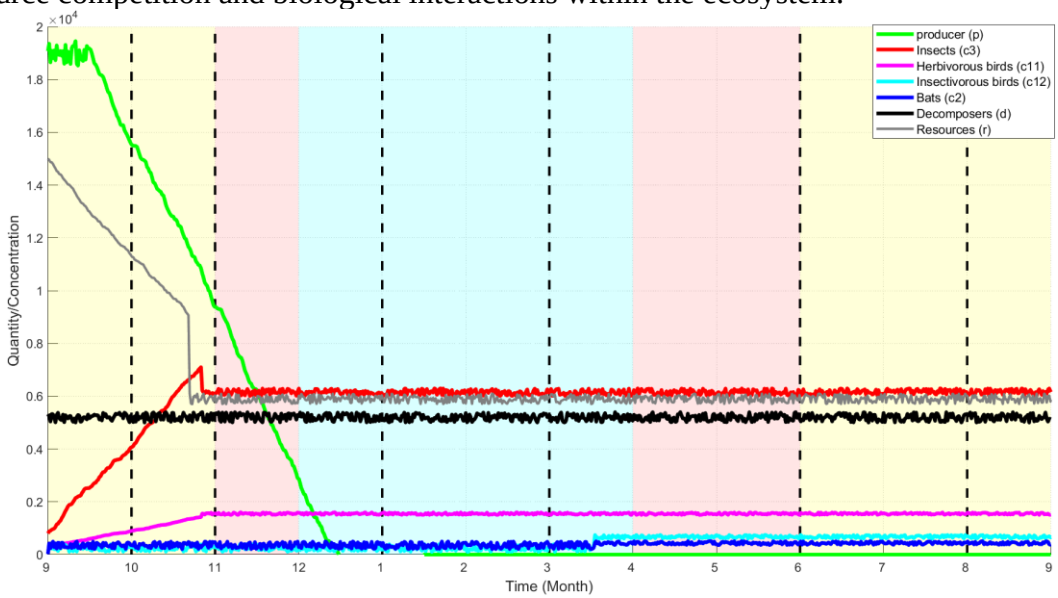


Figure 12 Result of impact of herbicide removal

This suggests that, with proper pest management, reducing herbicide use can promote ecosystem balance and sustainability.

7.2 Modeling the Impact of Bats

To more accurately simulate the impact of bats on producer reproduction in agricultural ecosystems, the original cellular automata model has been modified. While bats are included as consumers following the standard consumer rules, their role as pollinators is incorporated into the reproduction rules of producers. Bats, as key pollinators, have a significant positive effect on plant reproduction. Therefore, adjustments to the reproduction rules are necessary to reflect their role in promoting producer propagation.

7.2.1 Modifications to State Transition Rules

1. Modifications to Producer Reproduction Rules: Considering the pollination role of bats, the range and probability of producer reproduction are increased. The reproduction rules for producers are now divided into two categories: **Without bat coexistence**: These rules are covered in the previous model and are not reiterated here. **With bat coexistence**: This section focuses on the pollination effect of bats on producer reproduction.

When bats and producers coexist, the pollination effect of bats expands the reproduction range of producers $\xi_p^{'}$ and increases the reproduction probability. The modifications are as follows:

Step 1: Check if the energy is above the reproduction threshold, i.e., $S_4^t(i,j) > \theta_1$.

Step 2: Check if a random number exceeds the reproduction probability, i.e.,

$$\xi^{(t+1)}(i,j) > \xi_p^{'}.$$

Step 3: If conditions are met, select a random Moore neighbor for reproduction, with the expanded reproduction range $n, m \in [-p_0, p_0]$.

Step 4: After reproduction, the offspring inherit the parent's role and energy, and the parent's energy decreases. The reproduction probability is increased due to bat pollination:

$$S_1^{(t+1)}(i+n, j+m) = S_1^{(t)}(i,j)$$

$$S_4^{(t+1)}(i+n, j+m) = \alpha_1 S_4^{(t)}(i,j)$$

$$S_4^{(t+1)}(i+n, j+m) = (1-\alpha_1-\alpha_2) S_4^{(t)}(i,j)$$

Other rules remain unchanged. These modifications allow for a more precise simulation of the dual role of bats as insectivores and pollinators, reflecting their impact on ecosystem dynamics.

7.2.2 Model Solution and Analysis

Figure 13 illustrates the dynamic impact of bats on various components of the agricultural ecosystem.

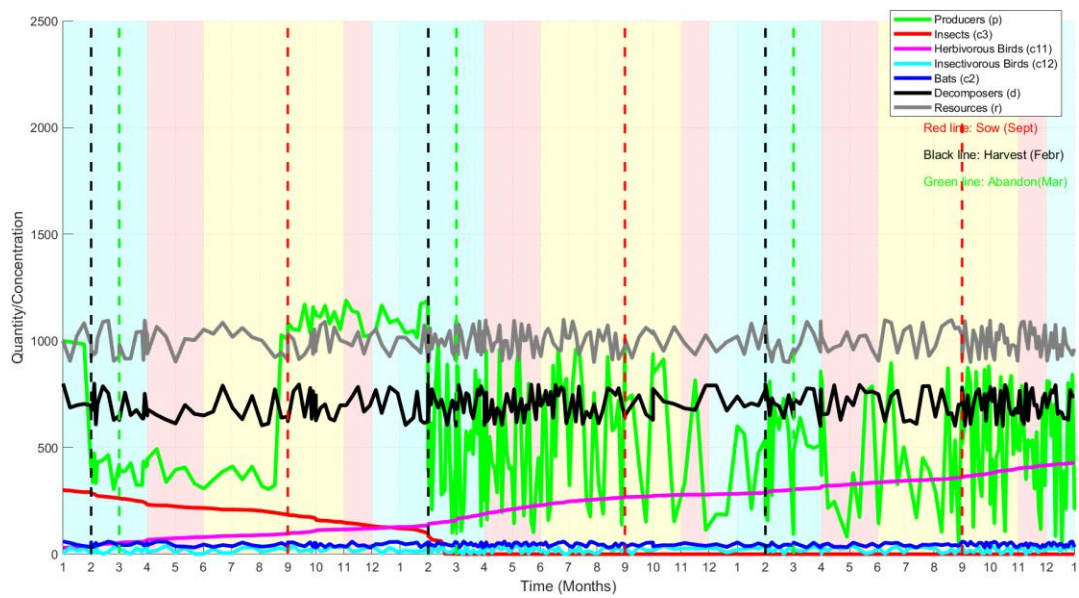


Figure 13 Result of impact of bats

The introduction of bats significantly reduces pest damage to crops, promotes the growth of producers, and improves ecosystem stability and resource utilization efficiency, demonstrating the important positive role of bats as a biological pest control measure in agricultural ecosystems.

7.3 Modeling the Impact of Rhizobia

In this study, **rhizobia** were selected as a beneficial species for detailed analysis. Rhizobia play a dual role in agricultural ecosystems: as decomposers, they contribute to organic matter decomposition and soil fertility maintenance; as symbionts, particularly with leguminous plants, they significantly enhance crop growth. Rhizobia form a symbiotic relationship with the roots of leguminous plants, catalyzing the conversion of atmospheric nitrogen into ammonia through nitrogenase activity, providing plants with essential nitrogen for growth. To incorporate this, the original cellular automata model was modified as follows.

7.3.1 Modifications to State Transition Rules

1. Modifications to Role Layer State Transition Rules: After introducing rhizobia, the relationship between producers and rhizobia in the model needed to be adjusted. As both decomposers and symbionts, rhizobia influence producer growth and reproduction. When plants coexist with rhizobia, their growth and reproduction are enhanced. Based on this relationship, the reproduction probability and range of producers increase when growing with rhizobia. The modification is as follows:

$$S_1^t(i,j) = p + d_1$$

$$S_1^t(i,j) = p + d_1 + t_2$$

2. Modifications to Resource Layer - Energy Layer Interaction Rules: When rhizobia form a symbiotic relationship with crops, the resource absorption and conversion rules need adjustment to reflect the nutritional support provided by rhizobia. With the introduction of rhizobia, the rules are modified as follows:

Step 1: Check if rhizobia and plants coexist in the grid, i.e., $S_3^t(i,j) = \{p_1, p_2, p_3\}$.

Step 2: If the condition is met, rhizobia provide nitrogen to enhance plant growth through nitrogen fixation, reducing the resource amount in the resource layer: $S_3^t(i,j) = S_3^t(i,j)\,(1-\alpha_3)$

Step 3: If producers are present, the absorbed resources are converted into the producer's energy, I_r a representing the nitrogen fixed by rhizobia: $S_4^t(i,j) = S_4^t(i,j) + \alpha_3 S_3^t(i,j) + I_r$

Other rules remain unchanged. These modifications allow the model to more accurately simulate the dual role of rhizobia as decomposers and symbionts, reflecting their impact on ecosystem dynamics.

7.3.2 Model Solution and Analysis

Figure 14 illustrates that the introduction of Rhizobia significantly promotes the growth of plants, stabilizes the population of decomposers, and enhances nutrient cycling. The gradual decrease in resource concentration over time indicates improved nutrient utilization efficiency.

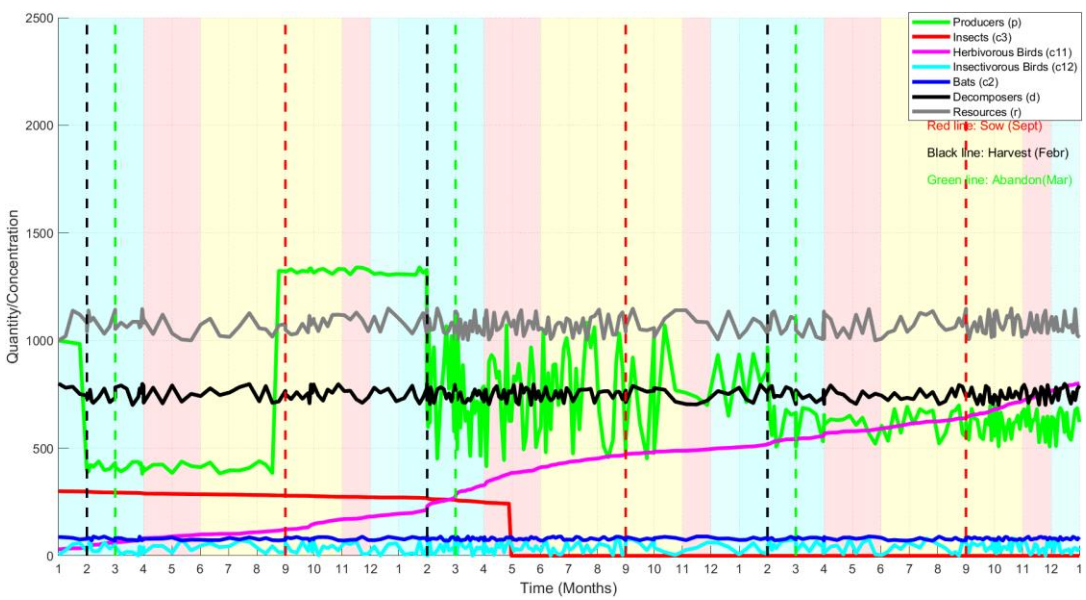


Figure 14 Result of impact of Rhizobia

Compared to bats, Rhizobia primarily support plant growth and nutrient cycling, while bats focus more on pest control and maintaining predator-prey balance. Together, they contribute to improved ecosystem stability and sustainability.

8 Task4: Modeling Organic Agriculture

8.1 Model Construction for the Organic Agriculture

To analyze the impact of farmers adopting organic farming methods, the model was further refined. Considering the variations within organic farming, two scenarios were introduced: **fully organic agriculture** and **basic organic agriculture**, as outlined in Table 5:

Table 5 classification of organic agriculture

Components	Fully Organic Agriculture	Basic Organic Agriculture
Fertilizers	Organic fertilizers	Organic fertilizers, small amounts of industrial fertilizers
Pest Control	Predatory species	Predatory species, small amounts of pesticides
Crop Planting Strategy	Diverse crops	Monoculture

Fully Organic Agriculture: $S_3^{(t+1)}(i,j) = S_3^{(t)}(i,j) + I_h^1$. Where I_h^1 represents the resource supplementation from organic fertilizers.

Basic Organic Agriculture: $S_3^{(t+1)}(i,j) = S_3^{(t)}(i,j) + I_h^2$ Where I_h^2 represents the resource supplementation from basic organic fertilizers.

Crop Strategy Considerations: Fully organic agriculture often utilizes mutually beneficial crops. To simplify the model, a symbiosis factor η was introduced to reflect the effect of diverse crops on resource absorption. The updated **resource-to-energy layer interaction rule** for fully organic farming is: $S_4^{(t+1)}(i,j) = \eta S_4^{(t+1)}(i,j)\,(1-\alpha_3)$

To quantitatively analyze pest control E_d ,crop health E_h ,biodiversity E_m and cost-effectiveness E_c across fully organic agriculture, basic organic agriculture, and conventional agriculture, the following indicators E_c were selected:

$$E_c = \frac{(\sigma E_m + B)}{H_i + H_o + H_p} \tag{10}$$

8.2 Model Solution and Analysis

Figure 15 was generated to compare the impact of different organic farming methods on the ecosystem and its components. The results show that **fully organic agriculture** has significant advantages in improving ecosystem stability and long-term sustainability.

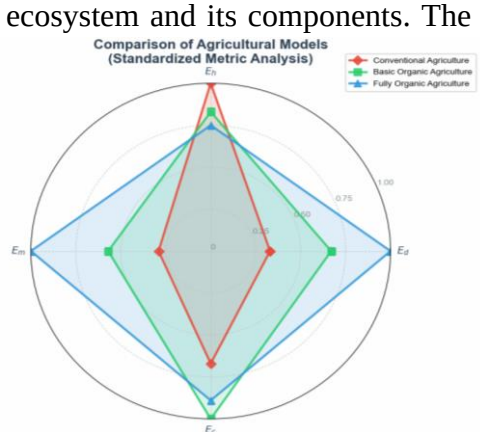


Figure 15 Radar chart of agriculture

- **Pest control**: Fully organic farming, through comprehensive biological control measures, reduces reliance on chemical pesticides, effectively managing pests while minimizing environmental pollution.
- **Crop health**: Conventional farming achieves the highest crop health indicators due to extensive chemical inputs, but this approach poses long-term risks to soil quality and human health.
- **Biodiversity**: Fully organic farming excels by promoting crop diversity and reducing pesticide use, enhancing ecosystem stability and resilience.

- **Cost-effectiveness**: Basic organic farming is a good transitional step, offering high short-term economic benefits while supporting ecological conservation. While fully organic farming has higher initial costs, such as introducing predators and improving soil, its long-term benefits can match or exceed those of basic organic farming.

In conclusion, fully organic farming boosts ecosystem sustainability and excels in areas like pest control and biodiversity, making it the ideal direction for modern agricultural development.

9 Sensitivity Analysis

This sensitivity analysis examines how changes in chemical concentrations affect bat, plant, and insect populations. By calculating the average populations of plants, insects, and bats at harvest time over 15 years under different chemical concentrations, we standardized the data to plot the relationship between chemical usage and population numbers.

In figure 16 , for plants, increasing chemical concentrations initially boost plant numbers, but high concentrations eventually slow growth, causing a decline. For insects, pesticide use reduces their numbers significantly, but some insects develop resistance, leading to a slow recovery as concentration rises. For bats, their population decreases as insect numbers fall, since they rely on insects for food.

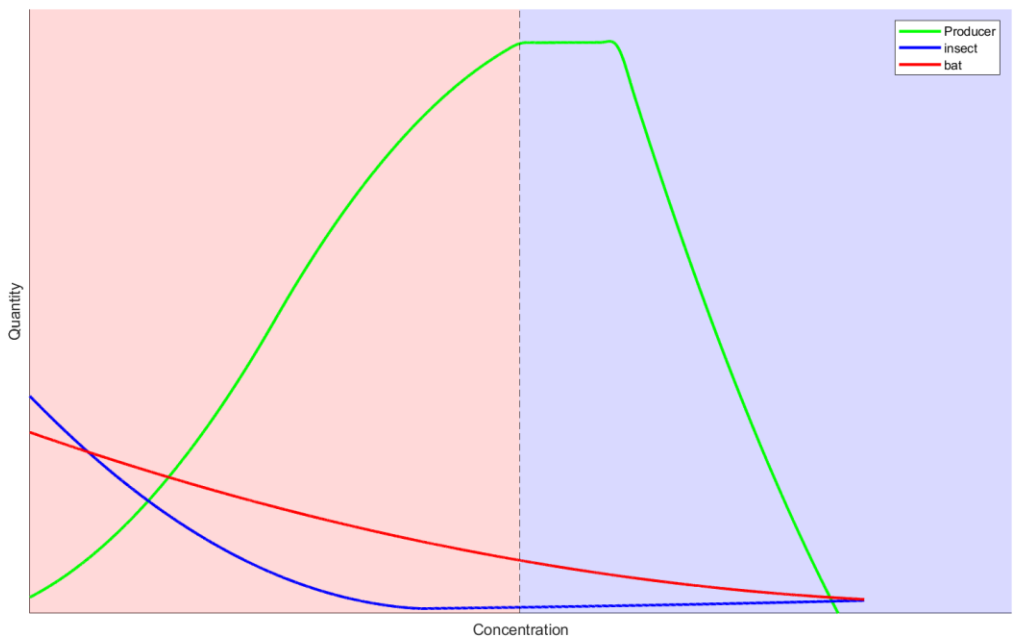


Figure 16 Sensitivity analysis

10 Model Evaluation

Strengths: The differential equation model accurately describes species dynamics, reveals internal relationships in food webs, and is supported by solid mathematical theory. Cellular automata models use simple, intuitive rules for state updates, making them easy to implement and suitable for simulating local interactions in ecosystems.

Weaknesses: The differential equation model has high computational costs and is less adaptable to complex ecosystems. Cellular automata models, limited to local interactions, struggle to handle long-range interactions. Both models have strengths and weaknesses suited to different scenarios.

Letter

Dear Farmer,

As President Xi Jinping said, *"Lucid waters and lush mountains are invaluable assets."* Organic farming embodies this principle, offering a sustainable path that protects the environment while ensuring economic prosperity. It is an eco-friendly agricultural model that preserves biodiversity, improves soil fertility, and meets the growing demand for healthier, eco-friendly food products.

The Benefits of Organic Farming

1. Sustainability: Organic farming reduces reliance on chemical fertilizers and pesticides, safeguarding water resources, improving soil health, and fostering biodiversity. By adopting natural inputs like compost and biological pest control, it creates a self-sustaining ecosystem while minimizing environmental pollution.

2. Economic Potential: While initial costs may be higher for organic fertilizers and pest control measures, organic products often fetch premium prices. Rising consumer demand for eco-friendly food makes organic farming a promising long-term investment. Healthier soils also reduce operational costs over time.

Strategies for Transition

1. Start Small: Begin with basic organic farming practices, such as using organic fertilizers and gradually reducing chemical inputs. Expand operations as you gain experience and resources.

2. Diversify Crops: Plant a variety of crops to enhance soil fertility, boost biodiversity, and increase resilience against pests and market changes.

3. Biological Pest Control: Use natural predators like bats, birds, and beneficial insects to control pests sustainably, reducing the need for chemical pesticides.

4. Collaborate Locally: Work with neighboring farmers to share resources and expertise, lowering costs and building a community of sustainable practices.

Economic and Policy Support

Transitioning to organic farming may require upfront investment, but government subsidies and programs often help offset costs. Explore support for organic certification, ecological projects, and financial assistance for organic inputs. Advocating for policies that encourage sustainable agriculture can further reduce expenses and enhance profitability.

While challenges exist, organic farming offers a balance between ecological conservation and economic growth. By adopting these practices, you can protect the environment, support biodiversity, and achieve long-term profitability.

Wishing you success in your organic farming journey!

Sincerely,

COMAP

Team # 2525028

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Report on Use of AI

1.Deepseek (Jan 26, 2025 version, Deepseek)

Query: How many Brazilian soybeans are ripe a year, when is the rainy season in the dry season, and when is the general planting time when to harvest

Output:

1)How many times do Brazilian soybeans ripen per year?

Brazilian soybeans are typically harvested once per year (one crop cycle). However, in some regions, farmers practice double cropping (e.g., planting corn or cotton after soybean harvest), but this does not mean soybeans ripen twice—they are only grown once annually.

2) Rainy Season vs. Dry Season

Rainy Season:

Central Brazil (e.g., Mato Grosso): September–May (peak rains December–February).

Southern Brazil (e.g., Paraná): October–March.

Dry Season:

Central Brazil: June–August (hot, minimal rain).

Southern Brazil: June–August (cooler, occasional frosts).

3) General Planting & Harvest Times

Planting:

Central Brazil: September–November (earliest start).

Southern Brazil: October–December (after wheat harvest).

Northeast Brazil: November–December.

Harvest:

Early crops: January (central regions).

Main harvest: February–March (nationwide peak).

Delayed harvest: April (southern regions).

Key Notes

Soybeans require rainy-season moisture for growth but face risks like disease (e.g., soybean rust) during heavy rains.

The dry season aids harvesting but may delay planting if rains arrive late.

Regional climate variations and government policies (e.g., sanitary void bans) impact exact timing.

2.Deepseek (Jan 26, 2025 version, Deepseek)

Query: Help me with the cellular automaton model

Output:

Building and Applying a Cellular Automaton Model

1.)Fundamental Concepts of Cellular Automaton (CA)

A Cellular Automaton is a discrete dynamical system composed of the following core components:

Grid: Typically a 2D lattice (e.g., checkerboard), but can also be 1D or 3D.

Cells: Individual units on the grid, each with finite states (e.g., 0/1, colors, species types).

Neighborhood: Defines the range of adjacent cells that influence a cell's state update. Common types:

Moore Neighborhood: Includes 8 surrounding cells (up, down, left, right, and diagonals).

von Neumann Neighborhood: Includes only 4 adjacent cells (up, down, left, right).

Rules: A function that determines the next state of a cell based on its current state and neighbor states.
2)Steps to Build a CA Model
Step 1: Define Model Parameters
Grid Size: E.g., a 100×100 2D grid.
Cell States: Examples:
Ecological model: Empty, Plant, Herbivore, Predator.
Epidemiological model: Susceptible, Infected, Recovered.
Neighborhood Type: Choose Moore or von Neumann based on the problem.
Step 2: Design State Transition Rules
Define how each state evolves based on neighbor interactions. Example rules for a predator-prey model:
Prey Reproduction: If adjacent empty cells exist, prey reproduce with probability
Predator Survival: Predators survive and reproduce if prey are nearby; otherwise, they die with probability
Empty Cells: Cells become empty after prey are consumed.
Step 3: Initialization and Boundary Conditions
Initialization: Set initial states (e.g., random distribution of prey and predators).
Boundary Conditions:
Fixed Boundaries: Edge cells remain static.
Periodic Boundaries: Grid edges wrap around (toroidal structure).
Step 4: Iteration and Visualization
Update cell states iteratively over discrete time steps.
Visualize the evolution using tools like Python's Matplotlib.
3)Implementation Tools and Code Example (Python)
Libraries:
NumPy: Efficient grid data handling.
Matplotlib: Visualization and animation.
Sample Code (Predator-Prey Model):
some Sample python code...
Applications and Extensions
Ecosystem Simulation: Species dispersal, resource competition, invasive species impact.
Epidemic Spread: Modeling disease transmission in populations.
Urban Planning: Traffic flow, fire propagation, land-use changes.
Physical Systems: Crystal growth, simplified fluid dynamics.
Advanced Extensions:
Multi-State Interactions: Introduce more species or environmental variables (e.g., temperature, resources).
Heterogeneous Grids: Different rules for distinct grid regions.
Parallel Computing: Accelerate large-scale simulations using GPU (e.g., CUDA).
gricultural productivity and sustainability.
5.)Common Issues and Solutions
Over-Simplified Rules: Calibrate parameters with real-world data to enhance realism.
Computational Inefficiency: Optimize neighbor-checking algorithms or use sparse grid representations.
Boundary Effects: Use periodic boundaries to minimize edge artifacts.